

CECS 277 – Lab 4 – File IO

State Capitals Quiz

Create a program that quizzes the user on the state capitals. Use the file 'statecapitals.txt' provided on Canvas to create the quiz questions.

Create the following functions:

1. `read_file_to_dict(file_name)` – each line in the file is in the format: `state, capital`. Read in each line, separate the state and the capital, and store them as a key:value pair in a dictionary. Return the filled dictionary.
2. `get_random_state(states)` – pass in the states dictionary. Convert the dictionary to a list of keys, then choose a random key from the list and return it.
3. `get_random_choices(states, correct_state)` – pass in the states dictionary and the correct state. Place the correct state into a list and then call `get_random_state` to add three other states to this list. These states should be different from the correct state and also different from each other. Using the list of states, create a list of capitals, shuffle it, and then return that list. This is the list of possible answers that the user will choose from.
4. `ask_question(correct_state, possible_answers)` – pass in the name of the correct state and the list of four possible answers. Display the question to the user and the four possible answers. Take in the user's selection and check that it is an A, B, C, or D. If it isn't, then display an 'Invalid' message and repeat until the user enters a valid choice. If it is, then convert the input to 0-3 (A=0, B=1, C=2, D=3) and return the value.

Your main should read in the file then have a loop that repeats 10 times, one for each of the quiz questions. For each quiz question, choose a random state (repeats are allowed) as the correct answer, generate the possible answers, display the question number and the quiz question. Compare the user's selection (0-3) against the location of the correct capital (0-3). If the user was correct, display a congratulatory message and give them a point, otherwise, tell them that they were incorrect and display the correct answer. After all 10 questions are finished, display the total points they received.

Partial Example Output (user input is in italics):

- State Capitals Quiz -

1. The capital of New Mexico is:

A. Phoenix B. Lansing C. Santa Fe D. Cheyenne

Enter selection: *G*

Invalid input. Input choice A-D.

Enter selection: *B*

Incorrect! The correct answer is: Santa Fe.

2. The capital of Florida is:

A. Tallahassee B. Dover C. Honolulu D. Sacramento

Enter selection: *A*

Correct!

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10. The capital of California is:

A. Hartford B. Sacramento C. Montgomery D. Juneau

Enter selection: *B*

Correct!

End of test. You got 8 correct.

Notes:

1. Place your name, date, and a brief description of the program in a comment block at the top of your program. Place brief comments throughout your code.
2. Use docstrings to document each of your functions.
3. Follow the guidelines in the Coding Standards reference document on Canvas.
4. You will need to use several list and dictionary functions, so it would be a good idea to review those lectures.
5. Use the random module to randomly choose from a list and to shuffle a list. You can review the reference document for random numbers on Canvas.
6. Do not create any extra functions or add any extra parameters or return values.
7. Please do not create any global variables, instead pass variables as arguments to the functions and return values back when needed.
8. You don't need the `check_input` module, but your `ask_question` function should be similar in that it always returns a valid input. It should repeatedly check the input until a valid input is entered.
9. Thoroughly test your program before submitting.
 - a. Make sure that you read in all 50 states and their capitals into the dictionary.
 - b. Make sure that you choose a random state for every question.
 - c. Make sure that your choices list always has 4 answers, one of them is the correct answer, and the others are random capitals that are not the correct one and are different from each other.
 - d. Make sure that the questions are correctly numbered.
 - e. Make sure that the question displays in the correct format.
 - f. Make sure that the user can only input A-D and it converts it properly into 0-3.
 - g. Make sure that the user's answer is checked properly.
 - h. Make sure that the user's points are counted properly.
 - i. Make sure the final total is correct.